



Tutorial 1 Mini-Lecture: Orbits & Formation

The concepts behind Worksheet 1 • Lectures 1–2

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What Worksheet 1 trains

Five problems, all built on Lectures 1–2. The physics you need, per problem:

1. **Weighing the Sun and Jupiter:** Kepler's third law as a scale.
2. **The cheapest route to Mars:** vis-viva and orbital energy.
3. **The Laplace resonance:** period ratios, conjunctions, tidal heating.
4. **Tides and the Roche limit:** differential gravity, reading a formula.
5. **Planetesimals to planets:** gravitational focusing and the growth regimes.

Every problem has the shape and level of one exam question. The worksheets define what the exam expects.

Weighing by orbits: Kepler's third law

Force balance on a circular orbit gives

$$M = \frac{4\pi^2 r^3}{GP^2}$$

- One period + one orbital size = the **central mass**.
- The orbiter's own mass **cancels**.
- Works for *any* central body: here, the Sun from Jupiter's orbit.

The fine print

The exact form contains $G(M + m)$: what you measure is the *sum* of both masses. For Jupiter about the Sun that is a ~ 0.1% effect.

Kepler's third law is a scale. Read carefully *what* it weighs.

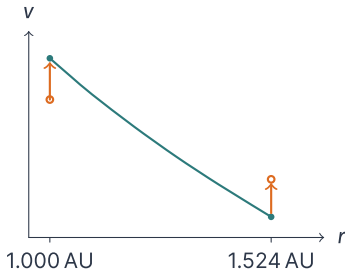
Vis-viva and orbital energy

Energy depends on a **alone**, and speed follows:

$$E = -\frac{GMm}{2a}, \quad \boxed{v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)}$$

- Circular orbit ($r = a$): $v = \sqrt{GM/a}$.
- Escape ($E = 0$): $v = \sqrt{2GM/r}$.
- Apsis distances: $r_p + r_a = 2a$.

An Earth–Mars transfer (curve) vs the circular speeds (circles); arrows mark the two speed increases.



Same a means same energy, whatever the eccentricity. Speeds then follow from vis-viva.

The Laplace resonance

- Io, Europa, Ganymede: $P = 1.769, 3.551, 7.155$ d, close to $1 : 2 : 4$; each pairwise ratio misses its integer slightly.
- The sharp statement uses **mean motions** $n = 2\pi/P$: a three-moon combination holds almost exactly.
- Io is always **opposite** at Europa–Ganymede conjunctions; no triple conjunction.

Why Io glows

The resonance pins $e \approx 0.004$. A locked moon on a *circular* orbit does not heat; the forced eccentricity keeps Io flexing: the most volcanically active body in the solar system.

Resonances act through repeated geometry: the same configuration returns, so small kicks add up.

Tides and the Roche limit

Tides are **differential** gravity, falling as d^{-3} :

$$\Delta a \approx \frac{2GM_p R_s}{d^3}$$

Inside the **Roche limit**, tides beat self-gravity and a body bound only by its own gravity is torn apart:

$$d_R \approx 2.46 R_p \left(\frac{\rho_p}{\rho_s} \right)^{1/3}$$

Read the formula before you use it

- The satellite's *size* cancels; only the **density ratio** matters.
- Saturn + solid ice: $d_R = 2.17 R_p$.
- Valid for bodies held by **self-gravity alone**; material strength is outside its domain.

Check limits and domain of a formula first; the numbers come second.

Planetesimals to planets

Gravitational focusing enlarges the cross-section:

$$\sigma_{\text{eff}} = \pi R^2 \left(1 + \frac{v_{\text{esc}}^2}{v_{\infty}^2} \right), \quad v_{\text{esc}} \propto R$$

- Cold swarm ($v_{\infty} \ll v_{\text{esc}}$): $\sigma_{\text{eff}} \propto R^4$, **runaway growth**.
- The winners stir the swarm, focusing switches off: **oligarchic growth**.

How big is the boost?

- Cold swarm ($v_{\infty} = 10$ m/s): a 500 km body focuses ~4000-fold.
- Hot swarm ($v_{\infty} = 500$ m/s): the boost collapses to a factor of a few.

Runaway growth ends itself; solids must lock in early.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** (“show that ...”) to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Try each problem alone *before* opening the solutions; they are complete, so reading first removes most of the benefit.

Worksheet and full solutions: ips.formingworlds.space/worksheets.html.
We discuss questions, not presentations of the solutions.